Polynomial Method in Combinatorics

Abisek Dewan & Rajiv Mishra

Indian Institute of Science Education and Research Kolkata, India

Objective

Our objective is to present the polynomial method and discuss the finite field Kakeya needle problem as an application.

Polynomial Methods

1. For a given set S, find a low degree polynomial over a finite field that vanishes on S or show that the only low degree polynomial that vanishes on S is the zero polynomial.
2. Find low dimensional polynomial space containing linearly independent polynomials corresponding to every element of the set S.

Kakeya Needle Problem

1. The Kakeya needle problem asks whether there is a minimum area of a region D in the plane, in which a needle of unit length can be turned through 360°.
2. A Kakeya set $K \subset \mathbb{R}^n$ is a set which contains a unit line segment in each direction.

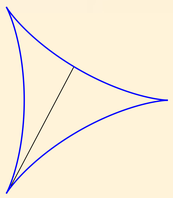


Figure 1:Unit Length Needle in Deltoid

Kakeya Conjecture for $\mathbb{R}^n$

For every $n \in \mathbb{N}$, a Kakeya set $K \subset \mathbb{R}^n$ has Hausdorff dimension n.

The Finite Field Kakeya Problem

- Let $\mathbb{F}^n$ be a vector space over a finite field $\mathbb{F}$.
- A Kakeya set $K \subset \mathbb{F}^n$ is a subset which contains a line in every direction, that is, for every $v \in \mathbb{F}^n$ there exists a $w \in \mathbb{F}^n$ such that the line $\{w + tv : t \in \mathbb{F}\}$ is contained in K.
- **Problem:** There exists a constant $c > 0$ depending only on the dimension n such that every Kakeya set $K \subset \mathbb{F}^n$ has cardinality $|K| \geq c|\mathbb{F}|^n$.

Dvir's Proof Using Polynomial Method

Theorem: Let $K \subset \mathbb{F}^n$ be Kakeya set. Then

$$|K| \geq \binom{|\mathbb{F}| + n - 1}{n} \geq \frac{|\mathbb{F}|^n}{n!}.$$

Proof

- Suppose $|K| < \binom{|\mathbb{F}|+n-1}{n}$.
- Then, $\exists$ a nonzero polynomial f of degree $\leq |\mathbb{F}| - 1$ such that $f(a) = 0$ for all $a \in K$.

  Reason: For $f(a_1, a_2, \ldots, a_n) = \sum_{\sum \alpha_i \leq |\mathbb{F}|-1} c_\alpha \, a_1^{\alpha_1} \cdots a_n^{\alpha_n} = 0$

  This is a system of equations in variables c_α.

  Note that number of variables $= \binom{|\mathbb{F}|+n-1}{n} > |K|$= number of equations.

  Therefore we have a nontrivial solution, thus such f exists.
- Let $f = p_0 + p_1 + \cdots + p_s$, where each p_i is a homogeneous polynomial of degree i.
- For any $v \in \mathbb{F}^n \setminus \{0\}$, there exists a $w \in \mathbb{F}^n \setminus \{0\}$ such that $w + tv \in K$ for every $t \in \mathbb{F}$.
- $R(t) := f(w + tv)$ is a polynomial of degree $s \leq |\mathbb{F}| - 1$ in t.
- As $R(t)$ is zero on $|\mathbb{F}|$ points (for $t \in \mathbb{F}$), $f(w + tv)$ is zero polynomial.

Proof (continues...)

- This implies $p_s(v) = 0$ for all $v \in \mathbb{F}^n \setminus \{0\}$.
- As p_s is the polynomial of degree $\leq |\mathbb{F}| - 1$, by the following Lemma, it has at most $(|\mathbb{F}| - 1)|\mathbb{F}|^{n-1} < |\mathbb{F}|^n - 1$ zeros, which is a contradiction. □

Lemma: Let $f(x_1, x_2, \ldots, x_n)$ be a polynomial over $\mathbb{F}$ of degree $\leq d$, then the number of zeros of f is at most $d|\mathbb{F}|^{n-1}$.

Remark: Best known bound is

$$|K| \geq \frac{|\mathbb{F}|^n}{2^n}.$$

Conclusion

- The Polynomial method is a relatively new algebraic technique that has been used to great effect to settle several long-standing open problems.
- We have seen the finite field Kakeya problem whose solutions invoke the polynomial method.
- In the past decade it enabled researchers to settle several long-standing open problems, few are as follows.
  1. Guth-Katz' solution of the Joints' Problem.
  2. The Cap-set problem and the work of Ellenberg-Gisjwijt.
  3. A function field analogue of Sárközy's theorem, due to Green.

References

[1] Larry Guth.
Polynomial methods in combinatorics, volume 64.
American Mathematical Soc., 2016.

[2] Stasys Jukna.
Extremal combinatorics: with applications in computer science, volume 571.
Springer, 2011.

[3] Zeev Dvir.
On the size of kakeya sets in finite fields.
Journal of the American Mathematical Society, 22(4):1093–1097, 2009.